

Spatio-Temporal Field-Modulated Fourier Neural Operator for Transmission Loss Estimation

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Abstract—Fourier Neural Operators (FNOs) are data-driven alternatives to conventional numerical solvers for rapid ocean acoustic transmission loss (TL) estimation. Recent work focuses on optimizing spatial network architectures for static environments and pays limited attention to temporal ocean-dynamic variability in acoustic fields. We propose the Spatio-Temporal Field-Modulated Fourier Neural Operator (STFM-FNO) to address this limitation. Instead of relying on straightforward spatio-temporal concatenation, STFM-FNO uses Feature-wise Linear Modulation (FiLM) to map Fourier-encoded time to block-wise affine parameters, inducing a spatio-temporally varying modulation field in latent feature maps without changing the spatial input. Under the proposed data-splitting strategy, STFM-FNO reduces estimation error relative to the evaluated static, concatenation-based, and conditional FNO baselines.

Index Terms—Fourier Neural Operator, Ocean Acoustics, Spatio-Temporal Field Modulation, Feature-wise Linear Modulation

I. INTRODUCTION

OCEAN acoustic transmission loss (TL) estimation is essential for downstream tasks such as environment-aware sensor placement optimization and autonomous vehicle path planning [1], [2]. Conventional methods perform these estimates by repeatedly running computationally expensive numerical solvers, such as the Range-dependent Acoustic Model (RAM) [3], under varying environmental configurations. These numerical approaches, while accurate, often struggle with real-time requirements [4]. To overcome this computational bottleneck, data-driven and physics-informed surrogate models have been explored for acoustic-field estimation and charting [5]–[7], fueled by broader successes of machine learning in acoustics [8]–[12], inverse problems [13], [14], and signal processing [15]–[17]. In particular, Fourier Neural Operators (FNOs) [18], [19] and their extensions [20]–[22] have become surrogate models for rapidly estimating TL fields from sound speed fields (SSFs). Recent variants have further integrated physics-informed constraints [23], [24] to improve generalization in acoustic charting, and applied them to broader wave propagation and seismic inversion problems [25]–[29].

Despite their efficiency, FNO applications in ocean acoustics largely assume a static ocean environment. In practice, ocean

sound-speed structure varies with temperature and salinity. An intuitive way to introduce temporal awareness is concatenating time t (as a scalar or encoding) with the input SSF along the channel dimension [18], [30]. However, direct scalar temporal concatenation provides only marginal change relative to the static baseline, whereas Fourier-feature concatenation may degrade performance (Sect. IV). Broadcasting a global temporal descriptor across the spatial grid may introduce spatially uniform, low-frequency signals that disturb the subtle spatial variations required for TL estimation.

We propose the Spatio-Temporal Field-Modulated Fourier Neural Operator (STFM-FNO), an architecture for TL estimation in time-varying ocean environments. The input consists of the SSF across range and depth, the locations of these points, and the time at which the SSF was collected. The output is TL at the same ranges and depths at that time. The time is used as an auxiliary conditioning variable to exploit the temporal organization of the environmental dataset. STFM-FNO moves beyond direct temporal concatenation and uses Feature-wise Linear Modulation (FiLM) [31] to inject temporal information into the FNO feature representation.

The time index is converted into a Fourier time encoding and then mapped to block-wise affine modulation parameters, which induce the latent modulation field $\mathcal{M}_\ell(\mathbf{x}, t)$ after interacting with spatial feature maps. By applying field modulation inside STFM-FNO blocks, STFM-FNO incorporates time information without directly broadcasting it into the SSF input. Numerical experiments using 3DSSF datasets [32] with RAM-computed TLs demonstrate that STFM-FNO consistently improves all reported metrics compared with the static and concatenation-based variants.

II. PROBLEM STATEMENT

A. Formulation for Time-Varying Ocean Acoustic Propagation

The complex-valued pressure field $p(r, z)$ at range r and depth z in an ocean waveguide is described by the 2-D Helmholtz equation under azimuthal symmetry [33]:

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r} \frac{\partial p}{\partial r} + \frac{\partial^2 p}{\partial z^2} + k_0^2 n^2(r, z)p = 0, \quad (1)$$

Here, $k_0 = \omega/c_0$ the reference wavenumber, ω the angular frequency, c_0 the reference sound speed (1500 m/s), and $n(r, z) = c_0/c(r, z)$ the refractive index.

For time-varying ocean conditions, the sound-speed field is written as $c_t(r, z)$, with refractive index $n_t(r, z) = c_0/c_t(r, z)$. Each time t therefore defines a corresponding acoustic propagation problem and TL field. With fixed source parameters

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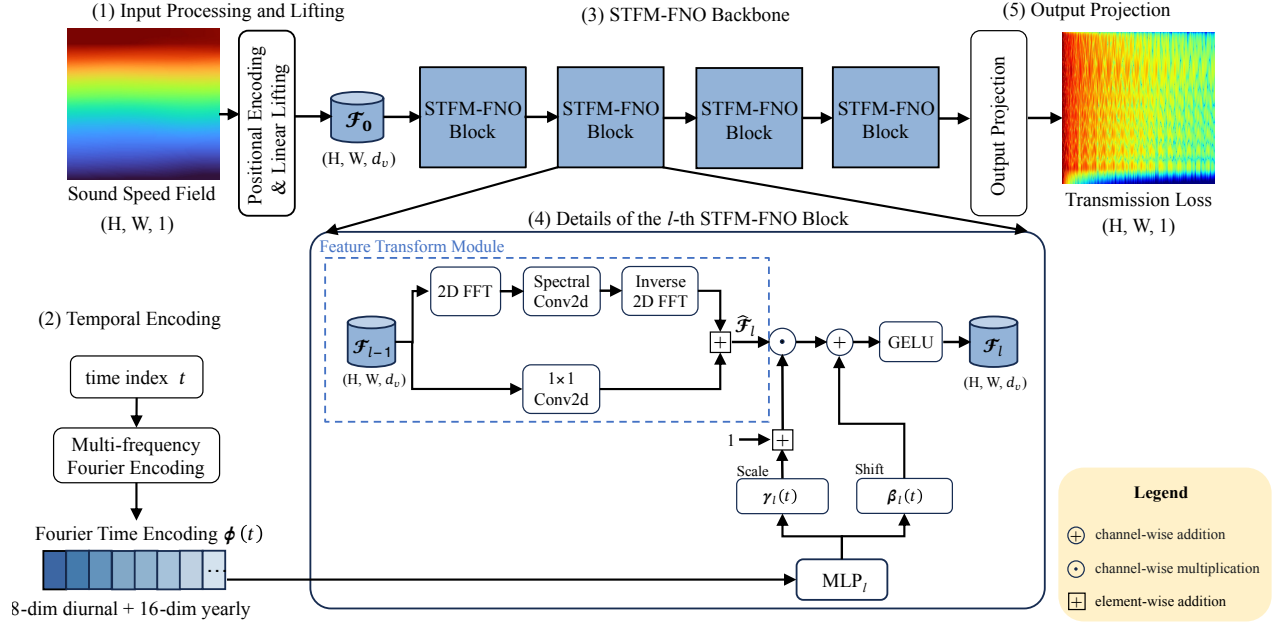


Fig. 1. Architecture of the Spatio-Temporal Field-Modulated Fourier Neural Operator (STFM-FNO). $\odot$ and $\oplus$ are channel-wise multiplication and addition, with one scalar per channel broadcast over the range-depth grid. Boxed plus is element-wise tensor addition. FFT: fast Fourier transform; MLP: multilayer perceptron; GELU: Gaussian error linear unit.

and boundary conditions, the current SSF determines TL through a static physical mapping $\mathcal{G}: \text{SSF} \mapsto \text{TL}$ [5], [18]. To exploit the temporal organization of the environmental dataset when approximating this mapping, we parameterize the learned surrogate as a family of operators $\{\mathcal{G}_t\}$ indexed by time:

$$\mathcal{G}_t : \text{SSF}_t \mapsto \text{TL}_t. \quad (2)$$

Here, time acts as an auxiliary conditioning variable that provides a temporal inductive bias for learning from finite data, rather than an additional physical state variable.

B. Limitation of Direct Concatenation

The FNO model [18] approximates the SSF-to-TL mapping without explicit temporal conditioning. The SSF channels vary over the range-depth grid, whereas the time channel has the same value at every grid point and contributes a spatially uniform, low-frequency signal. With direct concatenation, the initial FNO projection [18] from the concatenated input to latent channels must handle both spatial feature extraction and temporal conditioning.

III. PROPOSED METHOD

A. STFM-FNO Architecture

STFM-FNO uses two branches to incorporate temporal information into FNO-based TL estimation (Fig. 1). At each point (r, z) , the feature-processing branch concatenates $\text{SSF}_t(r, z)$ with normalized range $\bar{r}$ and depth $\bar{z}$, both in $[0, 1]$, giving $\mathbf{a}_t(r, z) = [\text{SSF}_t(r, z), \bar{r}, \bar{z}]^\top$.

A shared point-wise linear layer maps this vector to $\mathcal{F}_0(r, z) \in \mathbb{R}^{d_v}$. Over the full grid, this yields $\mathcal{F}_0 \in \mathbb{R}^{H \times W \times d_v}$, with H depth points, W range points, and $d_v = 64$ latent channels. The latent tensor passes through four STFM-FNO blocks.

In parallel, the temporal branch maps the time index t to a Fourier time encoding, which is further transformed into block-wise affine modulation parameters. These parameters are injected into each STFM-FNO block to induce a time-conditioned modulation field in latent feature maps. After the final block, the output projection uses two point-wise linear layers to produce one standardized TL value per grid point.

B. Multi-frequency Fourier Time Encoding

The Temporal Encoding module converts the time index t , associated with each SSF/TL pair, into the Fourier time encoding $\phi(t)$. Let $N_d = 8$ and $N_y = 365N_d = 2920$ be the daily and yearly sample counts. We decompose t into daily and yearly cycles and encode them using multi-frequency sine and cosine functions, where K_d and K_y are the numbers of sine-cosine frequency pairs used for the daily and yearly cycles, respectively. Let $\tau_d(t) = t \bmod N_d$ be the intra-day index:

$$\mathbf{f}_{\text{day}}(t) = [\sin(2\pi k \tau_d(t)/N_d), \cos(2\pi k \tau_d(t)/N_d)]_{k=1}^{K_d}, \quad (3)$$

$$\mathbf{f}_{\text{year}}(t) = \left[\sin\left(2\pi k \frac{t}{N_y}\right), \cos\left(2\pi k \frac{t}{N_y}\right) \right]_{k=1}^{K_y}. \quad (4)$$

The Fourier time encoding is then

$$\phi(t) = \begin{bmatrix} \mathbf{f}_{\text{day}}(t) \\ \mathbf{f}_{\text{year}}(t) \end{bmatrix}. \quad (5)$$

The daily and yearly periods provide physically motivated time scales for encoding recurrent environmental variability. Sine-cosine pairs represent temporal phase continuously across cycle boundaries, while multiple harmonics enrich the representation of within-cycle variations. This provides an explicit temporal inductive bias for learning from the one-year dataset [34].

Alternative temporal encodings are discussed in Sect. IV-C. Setting $K_d = 4$ and $K_y = 8$ gives $\phi(t) \in \mathbb{R}^{24}$, consisting of 8 daily and 16 yearly channels.

C. Details of an STFM-FNO Block

Within each STFM-FNO block, the two branches combine feature transformation with temporal affine modulation. This introduces temporal information directly at multiple feature-processing stages, whereas input concatenation supplies it before lifting.

In block ℓ , the Feature Transform Module applies parallel spectral and 1×1 pointwise convolutions to $\mathcal{F}_{\ell-1}$. Their sum gives $\widehat{\mathcal{F}}_\ell$, with both tensors in $\mathbb{R}^{H \times W \times d_v}$. The spectral and pointwise convolution parameters are shared across time, while temporal conditioning acts on the resulting features.

In the temporal branch, the Fourier time encoding $\phi(t)$ is passed to the block-wise multi-layer perceptrons (MLPs). For block ℓ , MLP_ℓ maps $\phi(t)$ to channel-wise FiLM modulation parameters [31]:

$$(\gamma_\ell(t), \beta_\ell(t)) = \text{MLP}_\ell(\phi(t)), \quad \gamma_\ell(t), \beta_\ell(t) \in \mathbb{R}^{d_v}. \quad (6)$$

Each block can therefore learn its own temporal response, adapting the conditioning to the features represented at that depth. The resulting affine parameters are applied to $\widehat{\mathcal{F}}_\ell$ to produce $\mathcal{F}_\ell$. The affine modulation is written through an additive latent modulation field:

$$\begin{aligned} \mathcal{M}_\ell(\mathbf{x}, t) &= \gamma_\ell(t) \odot \widehat{\mathcal{F}}_\ell(\mathbf{x}) + \beta_\ell(t), \\ \mathcal{F}_\ell(\mathbf{x}, t) &= \text{GELU}(\widehat{\mathcal{F}}_\ell(\mathbf{x}) + \mathcal{M}_\ell(\mathbf{x}, t)). \end{aligned} \quad (7)$$

Here, $\mathcal{M}_\ell(\mathbf{x}, t)$ is the latent spatio-temporal modulation field induced by the affine modulation. $\odot$ is channel-wise multiplication: $\gamma_\ell(t) \in \mathbb{R}^{d_v}$ is broadcast over the $H \times W$ range-depth grid and multiplies each channel of $\widehat{\mathcal{F}}_\ell$, whereas $\beta_\ell(t) \in \mathbb{R}^{d_v}$ is broadcast over the same grid and added channel-wise. Following the original FNO [18], we use the Gaussian Error Linear Unit (GELU) [35] as the block nonlinearity.

In the first block, $\widehat{\mathcal{F}}_\ell$ encodes the range-depth structure of the SSF; in deeper blocks, it also carries temporal information from preceding modulations. $\gamma_\ell(t)$ reweights feature channels, while $\beta_\ell(t)$ shifts their baseline levels to complement the multiplicative reweighting. Their time-conditioned affine action on spatially varying $\widehat{\mathcal{F}}_\ell(\mathbf{x})$ forms the latent spatio-temporal modulation field $\mathcal{M}_\ell(\mathbf{x}, t)$.

D. Training and Inference

During training, each batch contains SSF samples, time indices, and ground-truth TL fields. SSF inputs and TL targets are z-score standardized using training-set means and standard deviations. Let TL^n and $\widehat{\text{TL}}$ be the standardized target and estimate, respectively. With $\mathcal{R}(A, B) = \|A - B\|_2 / \|B\|_2$, we use a training objective that combines a field error term with first-order grid-difference error terms,

$$\begin{aligned} \mathcal{L}_A &= \mathcal{R}(\widehat{\text{TL}}^n, \text{TL}^n) + \beta_r \mathcal{R}(\Delta_r \widehat{\text{TL}}^n, \Delta_r \text{TL}^n) \\ &\quad + \beta_z \mathcal{R}(\Delta_z \widehat{\text{TL}}^n, \Delta_z \text{TL}^n), \end{aligned} \quad (8)$$

where Δ_r and Δ_z are first-order discrete grid differences along the range and depth directions, respectively. The directional weights β_r and β_z separately balance the range- and depth-difference penalties to account for the different range and depth discretizations. We set $\beta_r = \beta_z = 0.01$.

During inference, a given SSF and its time index are passed through the trained network once to produce $\widehat{\text{TL}}$, without running the RAM solver.

IV. EXPERIMENTS & RESULTS

A. Experimental Settings and Data-Splitting Strategy

We use the 3DSSF dataset of [32]. It contains 2920 SSF snapshots sampled eight times per day over one year. For each SSF, the corresponding TL field is generated using RAM [3] under the fixed shallow-water configuration in [32] (source frequency: 200 Hz; source depth: 50 m). Each SSF/TL pair is represented on a 199×800 range-depth grid, covering 0–40 km in range and 0–200 m in depth.

Random splitting can assign highly correlated neighboring SSFs to training and test sets, leading to overly optimistic performance estimates. We use a monthly data-splitting strategy with one-day purge neighborhoods. Each month contributes one central continuous test block. One-day intervals on either side of a test block are held out from training and testing, yielding 2144 training, 584 test and 192 purged samples.

The strategy retains seasonal coverage in the test set. Continuous test blocks and purge neighborhoods increase temporal separation between training and test observations, reducing the risk of temporal leakage caused by highly correlated neighboring samples. All SSF and TL normalization statistics are fitted on the training samples only.

We evaluate all models using Root Mean Square Error (RMSE), Relative L_2 Error (Rel. L_2), Structural Similarity Index Measure (SSIM), and Pearson Correlation Coefficient (PCC). $\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\widehat{\text{TL}}_i - \text{TL}_i)^2}$, where N is the number of range-depth grid points. RMSE and Rel. L_2 measure numerical error, whereas SSIM and PCC measure structural similarity and linear correlation between estimated and RAM-computed TL fields [5], [36].

For a controlled comparison, all variants use four Fourier blocks, $k_{\max, z} = 32$ and $k_{\max, r} = 128$ retained Fourier modes in the depth and range directions, and latent channel width $d_v = 64$. All models are trained for 100 epochs with Adam [37]. Baseline FNO uses the SSF and positional coordinates only. Concat (Scalar) broadcasts the normalized time index $t/(N_y - 1)$, where $N_y = 2920$, over the range-depth grid and concatenates it with the input before lifting. Concat (PE) broadcasts the 24-dimensional Fourier time encoding $\phi(t)$ over the grid and concatenates the resulting channels before lifting.

FNM-FNO uses an FNM-style vector-to-function decoder [38] to map the daily-and-yearly encoding to auxiliary spatial functions, whereas Low-rank Hyper-FNO uses a time-conditioned hypernetwork to generate low-rank updates to the Fourier kernels [39], [40]; their parameter counts are closely matched to STFM-FNO.

TABLE I
PERFORMANCE COMPARISON.

Model	RMSE (dB) ↓	Rel. L_2 ↓	SSIM ↑	PCC ↑
Baseline FNO	4.15	0.052	0.787	0.952
Concat (Scalar)	4.22	0.053	0.781	0.951
Concat (PE)	4.77	0.059	0.734	0.938
FNM-FNO	4.42	0.055	0.755	0.947
Low-rank Hyper-FNO	4.14	0.052	0.787	0.952
STFM-FNO (Ours)	3.95	0.049	0.800	0.956

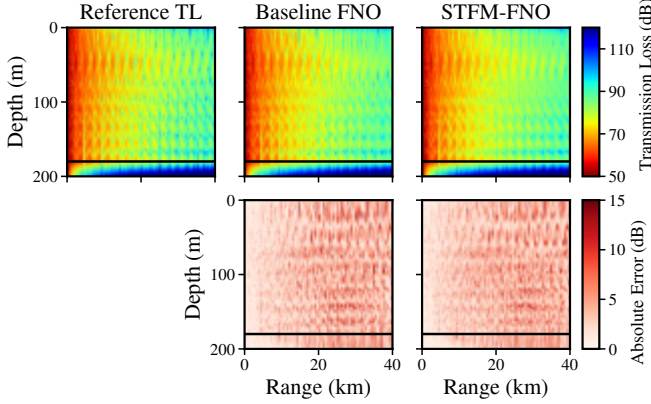


Fig. 2. TL estimates and absolute errors for a monthly test sample ($t = 1830$, Aug. 17); black line marks the 180-m bottom.

B. Performance Comparison

Table I shows that STFM-FNO performs best on all metrics, reducing RMSE by 0.20 dB (5%) versus Baseline FNO.

We assess cross-configuration robustness using a representative RAM configuration in which only the operating frequency is changed from 200 to 100 Hz. Both are retrained on the 100-Hz dataset using the settings in Sect. IV-A. They achieve RMSEs of 2.27 and 2.12 dB, respectively, yielding a 0.14-dB improvement. The advantage of STFM-FNO persists under this operating-frequency change.

C. Alternative Temporal Encodings

We conduct an ablation of temporal representations with the architecture and data-splitting strategy fixed. The alternatives address two questions: whether the daily and yearly scales improve on generic fixed frequencies, and whether learning frequencies is preferable to prescribing them. The RMSE is averaged over three runs.

Generic-Fourier-FiLM [34] uses frequency-agnostic sine-cosine features at 12 log-spaced frequencies without prescribing daily or yearly periods. Learned-frequency-FiLM [41] uses one learned linear channel and 23 sine channels with learned frequencies and phases.

The proposed encoding instead uses four daily and eight yearly sine-cosine harmonics. The three encodings yield comparable mean RMSEs (Table II).

D. Spatio-Temporal Field Modulation via Feature Interception

We evaluate the same held-out SSF at $t_{\text{winter}} = 360$ (Feb. 15) and $t_{\text{summer}} = 1800$ (Aug. 14), changing only the time index to examine its effect on feature modulation.

TABLE II
TEMPORAL-ENCODING ABLATION: RMSE (dB).

Temporal encoding	RMSE (dB) ↓
Generic-Fourier-FiLM	3.96
Learned-frequency-FiLM	3.91
Ours (daily + yearly)	3.90

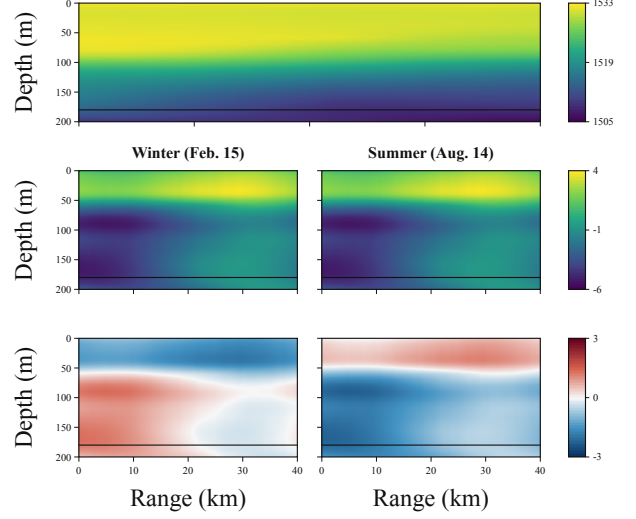


Fig. 3. Feature interception in the first STFM-FNO block. Rows show the input sound speed field (m/s), unmodulated feature $\hat{\mathcal{F}}_1$, and modulation field $\mathcal{M}_1$, respectively. Columns compare winter and summer conditioning for the same input.

We intercept the first STFM-FNO block after the Feature Transform Module and before affine modulation, recording the unmodulated intermediate feature $\hat{\mathcal{F}}_1(\mathbf{x})$, where $\mathbf{x}$ is a range-depth grid point. Using the corresponding affine parameters $\gamma_1(t)$ and $\beta_1(t)$, we compute the latent modulation field $\mathcal{M}_1(\mathbf{x}, t)$ (7), which is added to the unmodulated feature before GELU.

To show the temporal branch's direct effect, we inspect the first block because its intermediate features have not yet undergone temporal modulation. Keeping the SSF fixed separates this response from changes in the environmental input. Fig. 3 compares representative channels of $\hat{\mathcal{F}}_1$ and $\mathcal{M}_1$. The pre-modulation features remain similar, whereas $\mathcal{M}_1$ varies between the winter and summer runs, illustrating how time-conditioned affine parameters modulate latent spatial features.

V. CONCLUSION

The Spatio-Temporal Field-Modulated Fourier Neural Operator (STFM-FNO) conditions the Fourier neural operator (FNO) on time through block-wise spatio-temporal field modulation while retaining the sound speed field (SSF) as the physical input. It outperforms static, concatenation-based, and conditional FNO baselines across all metrics and evaluated configurations, and the three temporal encodings perform comparably. Feature interception confirms that the learned affine parameters modulate latent spatial features.

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